RESIDUE CLASSES AND STOPPING TIMES FOR AN ACCELERATED COLLATZ MAP

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Abstract

We introduce a modification of the Collatz function that shortens trajectories, reduces peak values, and suggests that the average stopping time remains bounded as the starting value grows. For a majority of starting values, a single step suffices to reach a strictly smaller integer. We provide probabilistic estimates based on a first-passage heuristic model, a linear composition formula, and detailed analyses for $n = 2^m - 1$ and $n = 10^m + j$. The principal rigorous result of this paper is an eventual stabilization theorem for the stopping times of integers of the form $10^m + j$. The remaining sections develop the arithmetic structure of the accelerated map and present numerical and heuristic evidence concerning starting values.

1. Introduction

The $3x+1$ problem or Collatz problem concerns the iteration of the Collatz function $C : \mathbb{N} \to \mathbb{N}$, defined as

$$C(x) = \begin{cases} 3x+1, & \text{if } x \equiv 1 \pmod 2; \\ x/2, & \text{if } x \equiv 0 \pmod 2. \end{cases}.$$

The classical conjecture is as follows.

Conjecture 1 (Collatz Conjecture)**.** Starting from any positive integer n, iterations of the function $C(n)$ will eventually reach the number 1 and thereafter cycle through the values $1, 4, 2, 1, \ldots$.

Terras [11] and Everett [3] introduced the compressed function

$$T(x) = \begin{cases} (3x+1)/2, & \text{if } x \equiv 1 \pmod 2; \\ x/2, & \text{if } x \equiv 0 \pmod 2, \end{cases}.$$

which combines two $C(n)$ steps whenever n is odd. The standard total stopping time for n under $C(n)$ is recorded in sequence A006577. Another variation is the Syracuse function $S(x)$, defined for odd x by

$$S(x) = (3x+1)/2^{\nu_2(3x+1)},$$

where $\nu_2(n)$ denotes the 2-adic valuation of n (the highest power of 2 dividing n).

Lagarias [5, 6, 7] provided comprehensive overviews of this research area.

The $3x+1$ conjecture has so far been verified computationally for all $n < 2^{71} \approx 2.3611 \cdot 10^{21}$ (Barina [1]).

This paper introduces a further modification of the Collatz function that produces shorter trajectories. We refer to it as the *accelerated Collatz function*. The approach taken here is based on the 2-adic structure of $n+1$ for odd n. Any odd integer n can be uniquely written as

$$n = r \cdot 2^k - 1, \quad \text{with } r \text{ odd and } k = \nu_2(n+1) \geq 1.$$

Applying $T(x)$ repeatedly k times to n yields

$$T^k(n) = 3^k r - 1.$$

Since r is odd and $k \geq 1$, the intermediate result $T^k(n)$ is an even number that can be divided by 2 one or more times. Setting $q = \nu_2(3^k r - 1)$, all of these divisions by 2 can be incorporated into a single step.

2. Definition and Basic Properties

We now give the central definition.

Definition 1. The accelerated Collatz function $\bar{c} : \mathbb{N} \to \mathbb{N}$ is defined by

$$\bar{c}(n) = \begin{cases} (3^k r - 1)/2^q, & \text{for odd } n = r \cdot 2^k - 1, \text{ with } k = \nu_2(n+1) \\ & \text{and } q = \nu_2(3^k r - 1); \\ n/2^k, & \text{for even } n = r \cdot 2^k, \text{ with } k = \nu_2(n). \end{cases}.$$

We let $\mathcal{L}(\mathcal{F}, n_0)$ denote the number of iterations required for the trajectory under a function $\mathcal{F}$. The initial value n_0 is not counted. It is sufficient to investigate the behavior until a smaller value $n_t < n_0$ is reached. Then n_t can be regarded as the new starting point until 1 is reached.

2.1. Basic Properties

1. *Always odd.* For any $n \in \mathbb{N}$, the value $\bar{c}(n)$ is always an odd integer because the denominator removes all factors of 2.

2. *Fixed point at 1.* Whereas $C(n)$ has the cycle $1 \to 4 \to 2 \to 1$ and $T(n)$ has the $1 \to 2 \to 1$ loop, the accelerated function satisfies $\bar{c}(1) = 1$ as a fixed point.

3. *Three out of four numbers stop in one step.* The definition preserves the property that each even number immediately leads to a smaller number. Furthermore, for $n \equiv 1 \pmod 4$, we have $k = 1$ and $n = 2r-1$, so $\bar{c}(r \cdot 2^1 - 1) = (3r - 1)/2^q \leq (3r - 1)/2 < 3/2\ r < n$. Thus, three out of four numbers stop after a single step, leaving only $n \equiv 3 \pmod 4$ for further investigation, excluding the trivial $n = 1$.

4. *Different first stopping values.*

   For $n_0 = 10087$, the trajectories for $C(n)$ and $T(n)$ both reach 13667 and eventually map to 7688, whereas $\bar{c}(n)$ maps 13667 directly to 961. Both 7688 and 961 are smaller than n_0, but not identical. This occurs because $\bar{c}(n)$ always applies the full divisor 2^q.

5. *Reduced maximum excursion.* For $n_0 = 27$, the maximal value in the $T(n)$ sequence is 3644, which reduces to 911 under $\bar{c}(n)$. Kontorovich and Lagarias [4] defined the maximum excursion ratio ρ as

$$\rho(n_0) = \max_{j \geq 1} \frac{\log T^j(n_0)}{\log n_0},$$

   which exceeds 2 when n_0^2 is exceeded. Of the eight numbers previously listed in A365482 exceeding $\rho > 2$ under $T(n)$ (with corresponding peak values listed in A365483), only three of them still have an excursion ratio greater than 2 under $\bar{c}(n)$, namely $n_0 = 27$, 3716509988199, and 1980976057694848447, the last of which was originally found by Oliveira e Silva [8].

   There are three even larger numbers with that property, under both $T(n)$ and $\bar{c}(n)$: 2641301410259797929, 2971464086542272671 and

   3961952115389696895 (all with 19 digits).

   The starting values giving rise to trajectory peaks without necessarily exceeding $\rho > 2$ are the path records A006884.

3. Acceleration

3.1. Examples

Starting with $n_0 = 27$, the trajectory for $C(n)$ takes 111 steps to reach 1 (A006577), and 96 steps to reach the smaller $n_t = 23$. With $T(n)$ it takes 59 steps to reach $n_t = 23$, while with $\bar{c}(n)$ only 15 steps suffice. This corresponds to an acceleration factor of approximately 3.9 relative to T, or 6.4 relative to C.

Below is the complete trajectory for $T(n)$, with the values calculated by $\bar{c}(n)$ highlighted in boldface:

27, 41, 62, **31**, 47, 71, 107, 161, 242, **121**, 182, **91**, 137, 206, **103**, 155, 233, 350, **175**, 263, 395, 593, 890, **445**, 668, 334, **167**, 251, 377, 566, **283**, 425, 638, **319**, 479, 719, 1079, 1619, 2429, 3644, 1822, **911**, 1367, 2051, 3077, 4616, 2308, 1154, **577**, 866, **433**, 650, **325**, 488, 244, 122, **61**, 92, 46, **23**.

Starting with $n_0 = 1023$, $C(n)$ takes 62 steps to reach 1 (29 to reach a smaller number), and $T(n)$ takes 18 steps. The function $\bar{c}(n)$ achieves this in only 2 steps: $1023, 7381, 173$.

3.2. Acceleration Factor

A single application of $\bar{c}(n)$ corresponds to k steps of $S(n)$, $k + q$ steps of $T(n)$, and $2k + q$ steps of $C(n)$.

The relevant expected values are summarized in the following proposition.

Proposition 1. *The expected values of k and q for $n \equiv 3 \pmod 4$ are given by*

$$\mathbb{E}(k \mid n \equiv 3 \pmod 8) = 2, \quad \mathbb{E}(k \mid n \equiv 7 \pmod 8) = 4, \quad \mathbb{E}(q) = 2.$$

Consequently, the average acceleration factor $F = 2k + q$ relative to $C(n)$ is 6 for $n \equiv 3 \bmod 8$ and 10 for $n \equiv 7 \bmod 8$.

Proof. For $n \equiv 3 \pmod 8$, $n + 1 = 4(2a + 1)$, so $k = \nu_2(n + 1) = 2$.

For $n \equiv 7 \pmod 8$, $n + 1 = 8(a + 1)$, so $k = 3 + \nu_2(a + 1)$. Since $\mathbb{P}(\nu_2(m) = j) = 2^{-(j+1)}$ for $j \geq 0$, in terms of asymptotic natural density, $\mathbb{E}(\nu_2(m)) = 1$. Thus, $\mathbb{E}(k \mid n \equiv 7 \pmod 8) = 3 + 1 = 4$.

For fixed k and $j \geq 1$, $3^k r \equiv 1 + 2^j \pmod{2^{j+1}}$ has a unique odd solution $r \pmod{2^{j+1}}$. Hence, q follows the geometric distribution $\mathbb{P}(q = j) = 2^{-j}$ for $j \geq 1$, and its expected value is $\mathbb{E}(q) = \sum_{j=1}^{\infty} j/2^j = 2$. □

4. Single Steps and Linear Composition

For fixed n, $\bar{c}(n)$ acts linearly as $\bar{c}(n) = A(k,q) \cdot n + c(k,q)$, where

$$A(k,q) = \frac{3^k}{2^{k+q}}, \qquad c(k,q) = \frac{3^k - 2^k}{2^{k+q}}.$$

The multiplier $A(k,q) > 1$ when $q \leq \lfloor \alpha k \rfloor$ with $\alpha = \log_2 3 - 1 \approx 0.58496$.

Accumulating t steps yields $n_t = (3^{K_t} n_0 + B_t)/2^{S_t}$, where

$$K_t = \sum_{i=0}^{t-1} k_i \quad , \quad Q_t = \sum_{i=0}^{t-1} q_i \quad , \quad S_t = K_t + Q_t,$$

and B_t is recursively defined by $B_0 = 0$ and $B_{i+1} = 3^{k_i} B_i + (3^{k_i} - 2^{k_i}) 2^{S_i}$. The recursive sequence B_t can be evaluated efficiently in $O(t)$ arithmetic operations via a Horner-like evaluation scheme during the step-by-step composition. From n_t as an odd integer, it follows that $3^{K_t} n_0 + B_t \equiv 2^{S_t} \pmod{2^{S_t+1}}$. Inverting 3^{K_t} specifies a unique start value residue class $n_0 \equiv 3^{-K_t}(2^{S_t} - B_t) \pmod{2^{S_t+1}}$. For example, $n_0 = 27$ reaching $n_t = 23$ gives $K_t = 37, Q_t = 22, S_t = 59$, and $B_t = 1100931843921811423$. Substitution into the composition formula gives $n_t = 23 < 27 = n_0$. Moreover, $Q_t = 22 > 37\alpha \approx 21.643$, so the cumulative multiplicative factor is less than 1.

5. Large Start Values and Trajectory Dynamics

5.1. One Step Suffices in Many Cases

For $n \equiv 3 \pmod 4$, a step decreases n when $q \geq \lfloor \alpha k \rfloor + 1$. The probability of an immediate reduction is

$$\mathbb{P}(n_1 < n_0) = \sum_{k=2}^{\infty} 2^{1-k} \cdot 2^{-\lfloor \alpha k \rfloor} \approx 0.42745.$$

This explicit single-step stopping probability corresponds to the exact summation with $\alpha = \log_2 3 - 1$, demonstrating that the immediate descent rate is strictly governed by the underlying 2-adic distribution.

Thus, roughly 42.7 percent of all starting values $n \equiv 3 \pmod 4$ result in a smaller number in the first step. This agrees with computational experiments, which give 42.745 percent for all $n_0 < 10^7$ for $n_0 \equiv 3 \pmod 4$.

5.2. Average Number of Steps for Large n

Numerical experiments indicate that the arithmetic mean of $\mathcal{L}(\bar{c}, n)$ taken for all $n < 10^m$, $n \equiv 3 \pmod 4$, approaches a constant close to 2.5 as m increases. Figure 1 displays the corresponding numerical means.

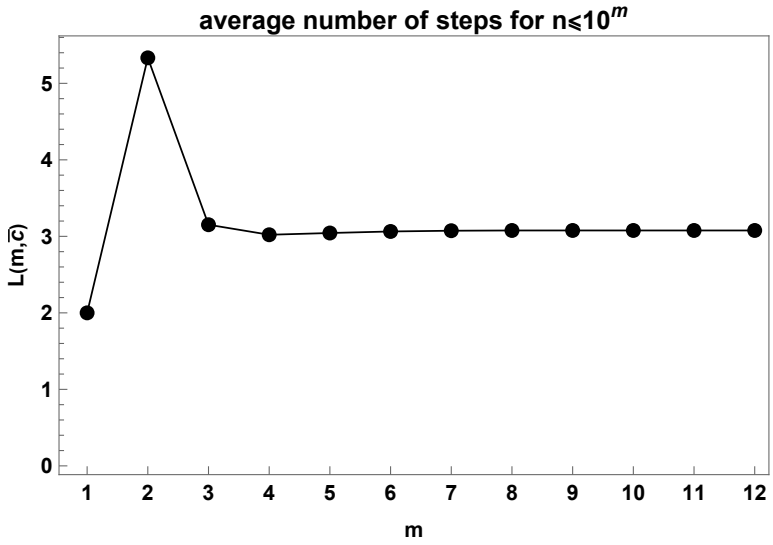


Figure 1: Average number of steps to reach a smaller number for $n < 10^m$, $n \equiv 3 \pmod 4$.

We use the following renewal approximation. After each non-stopping iteration, the residue data determining the next pair (k, q) are treated as newly sampled from the same asymptotic distribution and as independent of the preceding steps. If p denotes the probability that one iteration produces a value below the current reference value, then the corresponding first-passage time is modeled by a geometric random variable:

$$\mathbb{P}(t = s) = (1 - p)^{s-1} p, \qquad \mathbb{E}(t) = \frac{1}{p}.$$

This is a heuristic stochastic model; independence of successive Collatz iterates is not asserted.

For starting values $n \equiv 3 \pmod 8$, the model gives $p = 1/2$ and therefore

$$\mathbb{E}(t) \approx 2.$$

For starting values $n \equiv 7 \pmod 8$, the estimated continuation probability is

$$r = 1 - 4\sum_{k=3}^{\infty} 2^{-\lfloor k(\alpha+1) \rfloor} \approx 0.6451.$$

Thus, the corresponding stopping probability is $p = 1 - r \approx 0.3549$, giving

$$\mathbb{E}(t) \approx \frac{1}{0.3549} \approx 2.82.$$

The slight discrepancy between this heuristic estimate and the observed empirical mean (3.07692) for $n \equiv 7 \pmod 8$ arises from a weak correlation between consecutive iterates, which partially violate the idealized independence assumption.

Replacing r by the rough approximation $2/3$ gives the simpler estimate $\mathbb{E}(t) \approx 3$. The renewal approximation is intentionally simplified and does not reproduce the numerical means exactly. For $n < 10^{12}$ with $n \equiv 3 \pmod 4$, the observed arithmetic mean is 2.49266. The corresponding means in the residue classes 3 and 7 modulo 8 are 1.90838 and 3.07692, respectively.

5.3. Irregular Growth for $n = 2^m - 1$

For $n = 2^m - 1$, $k_1 = m$. By the lifting-the-exponent lemma [2, 10], $q_1 = 1$ for odd m, and $q_1 = 2 + \nu_2(m)$ for even m.

The residue classes $n \equiv -1 \pmod{2^m}$ force large initial growth $A(k, q)$, making these residue classes natural candidates for unusually long trajectories. This does not assert that these residue classes produce the globally longest trajectories.

Figure 2 compares these stopping times with the linear heuristic.

To estimate the number of iterations following the initial growth step, we use a simplified effective-drift model. We take $k_i = 2$, as in the residue class $n_i \equiv 3 \pmod 8$, and use the geometric valuation model $\mathbb{E}(q_i) = 2$. Since

$$\log h_i = k_i \log(3/2) - q_i \log 2,$$

this gives

$$\mathbb{E}[\log h_i] \approx 2\log(3/2) - 2\log 2 = \log(9/16).$$

This value is an effective heuristic drift, not an unconditional expectation over all odd Collatz iterates.

The logarithmic height gained in the first step is approximately

$$m \log(3/2) - q_1 \log 2.$$

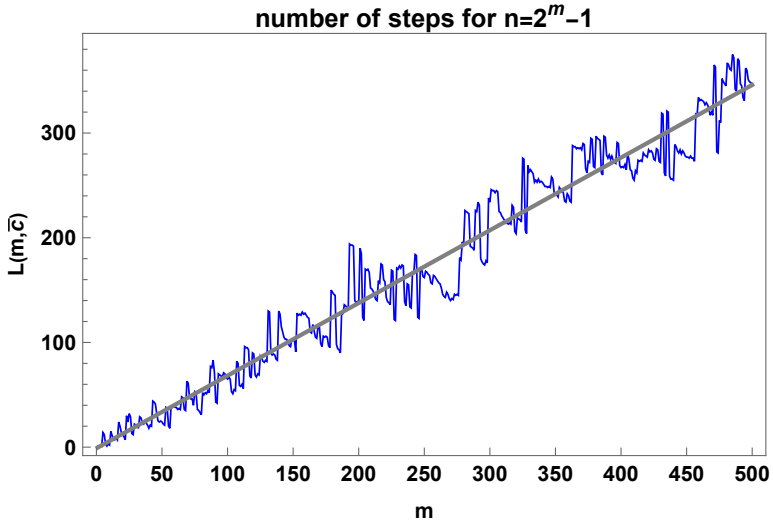


Figure 2: Number of steps for $n = 2^m - 1$, overlaid with $0.7047\,m$.

Dividing this gain by the magnitude of the assumed negative drift $\log(16/9)$ gives

$$\mathcal{L}(\bar{c}, 2^m - 1) \approx 1 + \frac{m \log(3/2) - q_1 \log 2}{\log(16/9)}.$$

Hence,

$$\mathcal{L}(\bar{c}, 2^m - 1) \approx 0.7047\, m - 1.2047\, q_1 + 1.$$

5.4. Bounded Growth for $n = 10^m + j$

We now give a sufficient condition under which $\mathcal{L}(\bar{c}, 10^m + j)$ is eventually constant. The result applies whenever the cumulative multiplier associated with the trajectory of j becomes smaller than 1.

Lemma 1. *Let j be a fixed odd integer, $n_0 = j$, $N_0 = 10^m + j$, and $S_t = \sum_{i=1}^{t}(k_i + q_i)$. If $m \geq S_t + 1$, then N_0 and n_0 share identical parameter pairs (k_i, q_i) for $1 \leq i \leq t$, and*

$$\bar{c}^t(10^m + j) = \frac{3^{K_t}}{2^{S_t}} 10^m + \bar{c}^t(j).$$

Proof. By induction on t. For $t = 0$, $N_0 \equiv n_0 \pmod{2^m}$. Assuming coincidence for i steps, $N_i - n_i = 3^{K_i} 2^{m - S_i} 5^m$, so $\nu_2(N_i - n_i) = m - S_i$. If

$m \geq S_t + 1$, then $m - S_i \geq k_{i+1} + q_{i+1} + 1$, preserving the parameter pair for step $i + 1$. □

We obtain the following consequence.

Corollary 1. *Let j be a fixed odd positive integer, and let (k_i, q_i) be the parameter pairs generated by the trajectory of j. Define*

$$K_t = \sum_{i=1}^{t} k_i, \qquad S_t = \sum_{i=1}^{t} (k_i + q_i), \qquad a_t = \frac{3^{K_t}}{2^{S_t}}.$$

Assume that the set $\{t \geq 1 : a_t < 1\}$ is nonempty, and define

$$\tau = \min\{t \geq 1 : a_t < 1\}.$$

Then there exists an integer $M(j)$ such that

$$\mathcal{L}(\bar{c}, 10^m + j) = \tau$$

for every $m \geq M(j)$.

Proof. Since a positive power of 3 cannot equal a positive power of 2, we have $a_i \neq 1$ for every $i \geq 1$. Hence, by the minimality of τ,

$$a_i > 1 \quad (1 \leq i < \tau), \qquad a_\tau < 1.$$

Choose m sufficiently large that Lemma 1 applies through step τ. Then

$$\bar{c}^i(10^m + j) - (10^m + j) = (a_i - 1)10^m + \big(\bar{c}^i(j) - j\big).$$

For each $i < \tau$, the coefficient $a_i - 1$ is positive. Hence, for all sufficiently large m,

$$\bar{c}^i(10^m + j) > 10^m + j.$$

At $i = \tau$, the coefficient $a_\tau - 1$ is negative, and therefore, for all sufficiently large m,

$$\bar{c}^\tau(10^m + j) < 10^m + j.$$

Thus, the first value below the initial value occurs after exactly τ iterations. □

The stabilization threshold can be determined explicitly in the following example.

Example 1. For $j = 31$, the parameter pairs along the first fourteen iterations give

$$K_{13} = 34, \qquad S_{13} = 53, \qquad \text{and} \qquad K_{14} = 35, \qquad S_{14} = 56.$$

Consequently,

$$a_{13} = \frac{3^{34}}{2^{53}} > 1, \qquad a_{14} = \frac{3^{35}}{2^{56}} < 1.$$

Thus, $\tau = 14$. Since $S_{14} = 56$, Lemma 1 and Corollary 1 imply

$$\mathcal{L}(\bar{c}, 10^m + 31) = 14 \qquad \text{for every } m \geq 57.$$

A direct calculation gives

$$\mathcal{L}(\bar{c}, 10^{55} + 31) = 15, \qquad \mathcal{L}(\bar{c}, 10^{56} + 31) = 14.$$

It follows that the exact stabilization threshold is $m^* = 56$.

Figure 3 illustrates this stabilization numerically.

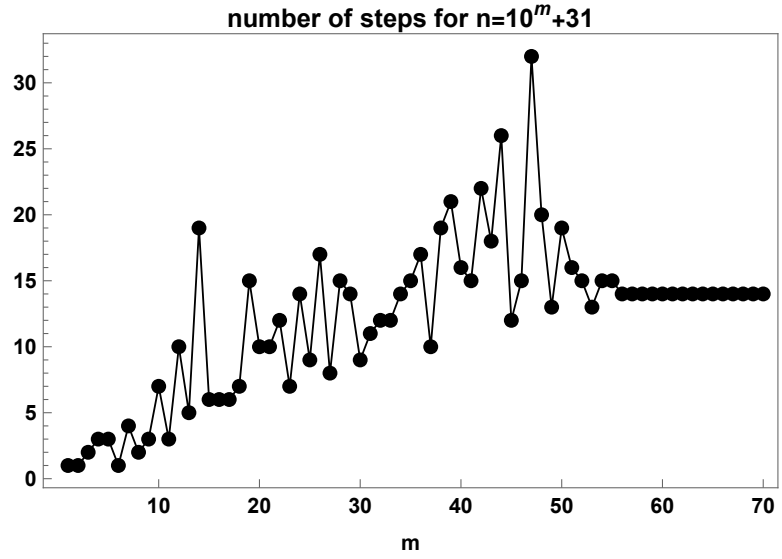


Figure 3: Number of steps for $n = 10^m + 31$, showing stabilization at 14 steps for $m \geq 56$.

6. Conclusion

The accelerated Collatz function $\bar{c}(n)$ provides a new tool for studying Collatz dynamics. Its use of the 2-adic structure of $n+1$ compresses several intermediate steps into a single iteration. We derive heuristic estimates for the families $n = 2^m - 1$ and $10^m + j$. Numerical experiments and heuristic arguments further suggest that the average stopping time over suitable residue classes may remain bounded.

7. List of OEIS Sequences

The sequences referenced in this paper are listed in the On-Line Encyclopedia of Integer Sequences [9]:

- A006577: Total stopping time under the Collatz function.
- A006884: Trajectory path records in the Collatz problem.
- A365482: Path records with maximum excursion ratio greater than 2.
- A365483: Corresponding maximum excursion values for A365482.

References

[1] D. Barina, Improved verification limit for the convergence of the Collatz conjecture, *J. Supercomput.* **81** (2025), 810.

[2] S. Cuellar and J. A. Samper, A nice and tricky lemma (lifting the exponent), *Math. Reflections* **3** (2007).

[3] C. J. Everett, Iteration of the number theoretic function $f(2n) = n, f(2n+1) = 3n+2$, *Adv. Math.* **25** (1977), 42–45.

[4] A. V. Kontorovich and J. C. Lagarias, Stochastic models for the $3x+1$ problem and generalizations, in J. C. Lagarias, ed., *The Ultimate Challenge: The* $3x+1$ *Problem*, Amer. Math. Soc., Providence, RI, 2010, 131–188.

[5] J. C. Lagarias, The $3x+1$ problem: an annotated bibliography (1963–1999), in J. C. Lagarias, ed., *The Ultimate Challenge: The* $3x+1$ *Problem*, Amer. Math. Soc., Providence, RI, 2010, 267–341.

[6] J. C. Lagarias, The $3x+1$ problem: an annotated bibliography, II (2000–2009), preprint, `arXiv:math/0608208`.

[7] J. C. Lagarias, The $3x+1$ problem: an overview, in J. C. Lagarias, ed., *The Ultimate Challenge: The* $3x+1$ *Problem*, Amer. Math. Soc., Providence, RI, 2010, 3–29.

[8] T. Oliveira e Silva, Empirical verification of the $3x+1$ and related conjectures, in J. C. Lagarias, ed., *The Ultimate Challenge: The* $3x+1$ *Problem*, Amer. Math. Soc., Providence, RI, 2010, 189–207.

[9] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, `https://oeis.org`.

[10] A. Parvardi, Lifting the exponent lemma (LTE), 2011. Available at `www.academia.edu/101947929/Lifting_The_Exponent_Lemma_LTE_Version_6`.

[11] R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976), 241–252.